

Spring & Spring Boot

Complete Topic-wise Interview Questions

Core Java • Spring Core • Spring Boot • Spring MVC • Spring Data JPA • Spring Security • Spring AOP • REST API • Testing •
Microservices • Docker • Design Patterns • SOLID • Hibernate • Collections • Multithreading • Exception Handling

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Questions only — no answers. Use for self-assessment, mock interviews, and revision.

Java Basics & Data Types

1. What is the difference between primitive types and reference types in Java?
2. What are the 8 primitive data types in Java? What are their sizes?
3. What is autoboxing and unboxing? What are the performance implications?
4. What is the difference between int and Integer in Java?
5. What is the difference between == and .equals() for String comparison?
6. What is String immutability? Why is String immutable in Java?
7. What is String pool (String interning)? How does it work in memory?
8. What is the difference between String, StringBuilder, and StringBuffer?
9. What is the difference between final, finally, and finalize in Java?
10. What is a static variable and static method? When would you use static?
11. What is the difference between static and instance methods?
12. What is a constructor? What is constructor chaining?
13. What is the difference between this and super keywords?
14. What is varargs in Java? How do you declare a varargs method?
15. What is the difference between local variable, instance variable, and class variable?
16. What is type casting? What is the difference between implicit and explicit casting?
17. What is instanceof operator and when do you use it?
18. What is the difference between break, continue, and return statements?

OOP Concepts

1. What are the 4 pillars of Object-Oriented Programming?
2. What is Encapsulation? How is it achieved in Java?
3. What is Inheritance? What is the difference between single and multiple inheritance?
4. What is Polymorphism? Explain compile-time vs runtime polymorphism.
5. What is Abstraction? What is the difference between abstract class and interface?
6. What is the difference between abstract class and interface in Java 8+?
7. Can an interface have a constructor? Can an abstract class have a constructor?
8. What is method overloading? What is method overriding?
9. What is the difference between method overloading and overriding?
10. What are the rules for method overriding (access modifier, return type, exceptions)?
11. What is the covariant return type in Java?
12. What is a default method in interface (Java 8)? Why was it introduced?
13. What is a static method in interface?
14. What is a functional interface? Give 5 examples from java.util.function.
15. What is the difference between Comparable and Comparator?
16. What is an inner class, static nested class, anonymous class, and local class?
17. What is object cloning? Difference between shallow copy and deep copy?
18. What is the Object class? What are its methods?

Java Collections Framework

1. What is the Collections Framework hierarchy? Draw the interface tree.

2. What is the difference between Collection and Collections?
3. What is the difference between ArrayList and LinkedList? When to use which?
4. What is the difference between ArrayList and Vector?
5. What is the internal working of HashMap in Java?
6. What is the difference between HashMap, LinkedHashMap, and TreeMap?
7. What is the difference between HashMap and Hashtable?
8. What is the difference between HashSet, LinkedHashSet, and TreeSet?
9. What happens when two keys have the same hashCode in HashMap? (hash collision)
10. What is the load factor and initial capacity of HashMap?
11. What is ConcurrentHashMap? How is it different from HashMap?
12. What is the difference between Iterator and ListIterator?
13. What is fail-fast vs fail-safe iterator?
14. What is the difference between Comparable and Comparator? How to sort a custom object?
15. What is Collections.sort() vs Arrays.sort()? What algorithm does each use?
16. What is the difference between Stack and Queue?
17. What is PriorityQueue in Java? How does it order elements?
18. What is Deque? When would you use ArrayDeque?
19. What is the difference between List.of() and new ArrayList() in Java 9+?
20. What is an unmodifiable collection? How do you create one?
21. What is the difference between remove() and removeIf() in collections?

Streams, Lambdas & Functional Programming

1. What is a Lambda expression? What is the syntax?
2. What is the difference between a lambda and an anonymous class?
3. What is a Stream in Java? How is it different from a Collection?
4. What is the difference between intermediate and terminal operations in Streams?
5. What are lazy operations in Streams? Why are they important for performance?
6. Explain filter(), map(), flatMap(), sorted(), distinct(), limit(), skip().
7. Explain collect(), count(), findFirst(), findAny(), anyMatch(), allMatch(), noneMatch().
8. What is reduce()? Give an example of summing a list of integers.
9. What is Collectors.groupingBy()? What is Collectors.partitioningBy()?
10. What is the difference between map() and flatMap()?
11. What is Optional? Why was it introduced? How is it used?
12. What is method reference? What are the 4 types of method references?
13. What is a Predicate, Function, Consumer, Supplier, UnaryOperator, BinaryOperator?
14. What is the difference between Stream.of() and Arrays.stream()?
15. What is a parallel stream? When should you use it and what are the risks?
16. What is Stream.peek()? When is it used?

Exception Handling

1. What is the exception hierarchy in Java? What is Throwable?
2. What is the difference between checked and unchecked exceptions?
3. What is Error vs Exception? When is an Error thrown?
4. What is the difference between throw and throws?
5. What is a try-with-resources statement? How does it work?
6. What is a multi-catch block (Java 7)? What is its syntax?
7. Can you have a try block without a catch block? Can you have finally without catch?
8. What happens if an exception is thrown in the finally block?
9. What is exception chaining? How do you preserve the original exception?
10. How do you create a custom exception? When should it be checked vs unchecked?
11. What is NullPointerException? How do you prevent it?
12. What is StackOverflowError? What causes it?
13. What is OutOfMemoryError? What are its common causes?
14. What is ClassCastException? When does it occur?
15. What is ConcurrentModificationException? How do you avoid it?
16. What is the difference between NoSuchMethodException and NoSuchMethodError?

Generics

1. What are Generics in Java? Why were they introduced?
2. What is type erasure in Generics?
3. What is the difference between a generic class and a generic method?
4. What is an upper-bounded wildcard (? extends T)? Give an example.
5. What is a lower-bounded wildcard (? super T)? Give an example.
6. What is the PECS principle (Producer Extends, Consumer Super)?
7. What is the difference between List and List?
8. Can you create an instance of a generic type T inside a generic class?
9. What is a bounded type parameter? How is it declared?
10. What is a raw type and why is it dangerous?

Multithreading & Concurrency

1. What is a Thread? What is the difference between process and thread?
2. What are the ways to create a thread in Java?
3. What is the thread lifecycle? Name all 6 states.
4. What is the difference between start() and run() methods of Thread?
5. What is synchronized keyword? What is a synchronized block vs synchronized method?
6. What is the difference between volatile and synchronized?
7. What is a deadlock? How do you detect and prevent it?
8. What is a race condition? Give an example.
9. What is the happens-before relationship in Java memory model?
10. What is wait(), notify(), notifyAll()? Why must they be called from synchronized block?
11. What is the difference between sleep() and wait()?

12. What is a thread pool? What is `Executors.newFixedThreadPool()`?
13. What is `ExecutorService`? What is the difference between `submit()` and `execute()`?
14. What is `Future` and `Callable`? How are they different from `Runnable`?
15. What is `CompletableFuture`? What is its advantage over `Future`?
16. What is a `ReentrantLock`? How is it different from `synchronized`?
17. What is a `Semaphore` in Java concurrency?
18. What is the difference between `CopyOnWriteArrayList` and `ArrayList` in concurrency?
19. What is `AtomicInteger`? When would you use atomic classes?

IoC Container & ApplicationContext

1. What is the Spring Framework? What problem does it solve?
2. What is Inversion of Control (IoC)? How is it different from traditional programming?
3. What is the difference between BeanFactory and ApplicationContext?
4. What are the different types of ApplicationContext implementations?
5. What is a Spring Bean? Who creates and manages it?
6. How does the ApplicationContext load bean definitions?
7. What is the Spring IoC container initialisation process?
8. What is the difference between an XML-based, annotation-based, and Java-based Spring configuration?
9. What is a @Configuration class? How is it different from a regular @Component?
10. What is component scanning? How do you enable it?
11. What is the difference between @ComponentScan and @EnableAutoConfiguration?
12. Can you have multiple ApplicationContext instances in one application?
13. What is a parent-child ApplicationContext relationship?
14. What is ApplicationContextAware and BeanFactoryAware?

Dependency Injection

1. What is Dependency Injection? What are the benefits?
2. What are the 3 types of Dependency Injection in Spring? Which is recommended and why?
3. What is constructor injection and why does Spring team recommend it?
4. What is field injection and why is it not recommended for production code?
5. What is setter injection and when would you prefer it over constructor injection?
6. What is @Autowired? How does Spring resolve it?
7. What happens if two beans of the same type exist and you @Autowired by type?
8. What is @Qualifier? How is it used with @Autowired?
9. What is @Primary? How does it differ from @Qualifier?
10. What is circular dependency? How does Spring handle it? How do you fix it?
11. What is @Lazy? How does it help with circular dependency?
12. What is @Bean? How is it different from @Component?
13. What is the difference between @Component, @Service, @Repository, @Controller?
14. What is @Value? How do you inject a property from application.properties?
15. What is @ConfigurationProperties? How is it different from @Value?
16. What is Spring Expression Language (SpEL)? Give an example of @Value with SpEL.
17. What is a stereotype annotation? What does it do?

Bean Lifecycle & Scopes

1. What is the complete Spring Bean lifecycle from instantiation to destruction?
2. What is @PostConstruct and @PreDestroy? At which point in the lifecycle do they run?
3. What is InitializingBean and DisposableBean? How are they different from @PostConstruct?
4. What is BeanPostProcessor? What two methods does it provide?
5. What is BeanFactoryPostProcessor? When does it run in the lifecycle?
6. What are the Spring Bean scopes? Name all of them.

7. What is Singleton scope? Is it thread-safe? Why?
8. What is Prototype scope? When would you use it?
9. What are web-aware scopes: Request, Session, Application, Websocket?
10. What happens when a Singleton bean depends on a Prototype bean?
11. How do you inject a Prototype bean into a Singleton bean correctly?
12. What is a scoped proxy? How is it created?
13. What is bean definition inheritance in Spring?
14. What is a FactoryBean? How is it different from a regular bean?
15. What is @DependsOn? When would you use it?
16. What is @Scope annotation? How do you create a custom scope?

Spring Boot Core Concepts

1. What is Spring Boot? What are its advantages over plain Spring?
2. What is convention over configuration? How does Spring Boot implement it?
3. What is `@SpringBootApplication`? What 3 annotations does it combine?
4. What is the entry point of a Spring Boot application? How does `SpringApplication.run()` work?
5. What is the difference between Spring Boot 2.x and Spring Boot 3.x?
6. What is the embedded server in Spring Boot? How do you change from Tomcat to Jetty?
7. What is a fat JAR (uber JAR)? How does Spring Boot package it?
8. How do you run a Spring Boot application from command line?
9. What is Spring Boot CLI? When would you use it?
10. What is `@EnableAutoConfiguration`? How does the auto-configuration mechanism work?
11. What is `spring.factories` file (Spring Boot 2) and `AutoConfiguration.imports` (Spring Boot 3)?
12. What is `@Conditional`? What are its variants: `@ConditionalOnClass`, `@ConditionalOnMissingBean`?
13. What is `@ConditionalOnProperty`? Give an example.
14. How do you exclude a specific auto-configuration class?
15. What is `@SpringBootTest`? What does it load?

Application Properties & Configuration

1. What is `application.properties` vs `application.yml`? Which do you prefer and why?
2. What is the order of precedence for Spring Boot property sources?
3. How do you externalise configuration in Spring Boot?
4. What is `@Value`? What is `@ConfigurationProperties`? When to use which?
5. What is a relaxed binding in `@ConfigurationProperties`?
6. How do you define and use profiles in Spring Boot?
7. What is `application-dev.properties`? How does it override `application.properties`?
8. How do you activate a Spring profile — 3 different ways?
9. What is `spring.config.location` and `spring.config.name`?
10. How do you pass properties at runtime using command-line arguments?
11. What is environment variable injection? How do you override a property with an env var?
12. What is `@Profile`? How do you create environment-specific beans?
13. What is a Spring Boot configuration server? What is Spring Cloud Config?
14. What is `@PropertySource`? How do you load a custom properties file?
15. What is the difference between `spring.profiles.active` and `spring.profiles.include`?

Spring Boot Starters & Auto-Configuration

1. What is a Spring Boot Starter? How is it structured?
2. What does `spring-boot-starter-web` include?
3. What does `spring-boot-starter-data-jpa` include?
4. What does `spring-boot-starter-security` include?
5. What does `spring-boot-starter-test` include?
6. What does `spring-boot-starter-actuator` include?
7. How do you create a custom Spring Boot Starter?

8. What is spring-boot-starter-validation? What annotations does it enable?
9. What is spring-boot-starter-mail? What does it auto-configure?
10. What is spring-boot-starter-cache? What caching providers does it support?
11. What is spring-boot-devtools? What features does it provide?
12. What is the difference between spring-boot-devtools restart and hot reload?
13. What is the spring-boot-maven-plugin? What goals does it provide?

Spring Boot Actuator

1. What is Spring Boot Actuator? What is its purpose?
2. What are the built-in Actuator endpoints? Name at least 10.
3. What is the /actuator/health endpoint? How do you customise it?
4. What is the /actuator/info endpoint? How do you populate it?
5. What is the /actuator/metrics endpoint? What metrics does it expose?
6. What is the /actuator/env endpoint? What does it show?
7. What is the /actuator/beans endpoint? What does it expose?
8. What is the /actuator/mappings endpoint?
9. How do you expose or secure Actuator endpoints?
10. What is management.endpoints.web.exposure.include=*?
11. What is a custom HealthIndicator? How do you create one?
12. What is Micrometer? How does it integrate with Actuator?
13. What is the /actuator/loggers endpoint? What can you do with it?
14. What is the /actuator/threaddump endpoint used for?
15. How do you change the Actuator base path from /actuator to something else?

Spring MVC Core

1. What is the DispatcherServlet? What role does it play in Spring MVC?
2. Explain the complete Spring MVC request processing lifecycle.
3. What is a HandlerMapping? Name 3 implementations.
4. What is a HandlerAdapter? What does it do?
5. What is a ViewResolver? Name 3 implementations.
6. What is ModelAndView? When would you return it from a controller?
7. What is the difference between @Controller and @RestController?
8. What is @ResponseBody? What does it do to the return value of a controller method?
9. What is HttpMessageConverter? Name 3 built-in converters.
10. How does Spring MVC choose which HttpMessageConverter to use?
11. What is content negotiation in Spring MVC?
12. What is @RequestMapping? What attributes does it have?
13. What is the difference between @GetMapping, @PostMapping, @PutMapping, @PatchMapping, @DeleteMapping?
14. What is @PathVariable? How do you use it with regular expressions?
15. What is @RequestParam? What are its required, defaultValue, and name attributes?
16. What is @RequestBody? What is @ResponseBody? How do they work with Jackson?
17. What is @RequestHeader? Give an example use case.
18. What is @CookieValue? How do you read a cookie in a Spring controller?
19. What is @ModelAttribute? How is it used in form-based controllers?
20. What is @SessionAttribute and @SessionAttributes?

REST API Design & HTTP

1. What are the HTTP methods? What are their semantics (GET, POST, PUT, PATCH, DELETE)?
2. What is REST? What are its 6 architectural constraints?
3. What is idempotency? Which HTTP methods are idempotent?
4. What are safe HTTP methods?
5. What are the HTTP 2xx, 3xx, 4xx, 5xx status codes? Name the most important ones.
6. What is the correct HTTP status code when a resource is created (POST)?
7. What is the correct HTTP status code when a resource is not found?
8. What is the difference between PUT and PATCH?
9. What is ResponseEntity? How is it used to control HTTP status codes?
10. What is @ResponseStatus? How does it work?
11. What is UriComponentsBuilder? How do you return a Location header after POST?
12. What is HATEOAS? What is spring-hateoas?
13. What is API versioning? What are the different strategies?
14. What is a REST resource? How do you design resource URLs?
15. What is the difference between /api/v1/users/{id} and /api/v1/users?id={id}?
16. What is idiomatic REST — what makes an API truly RESTful?

Request Validation & Error Handling

1. What is Bean Validation (JSR-380)? How do you enable it in Spring Boot?
2. What are the common Bean Validation annotations?
3. What is @Valid vs @Validated? What is the difference?
4. What is MethodArgumentNotValidException? How do you handle it globally?
5. What is ConstraintViolationException? When does it differ from MethodArgumentNotValidException?
6. What is @ExceptionHandler? How does it work?
7. What is @ControllerAdvice vs @RestControllerAdvice?
8. How do you return a structured error response (timestamp, status, message, path)?
9. What is ProblemDetail (RFC 7807)? How does Spring 6 support it?
10. What is @Valid on method parameters vs class-level? What is groups in Bean Validation?
11. How do you create a custom constraint annotation?
12. What is HandlerExceptionResolver? Name 3 implementations.
13. What is DefaultHandlerExceptionResolver?
14. What is @CrossOrigin? How does it differ from global CORS configuration?
15. What is CORS? Why do browsers enforce it? How do you configure it in Spring Boot?

JPA Entities & Mappings

1. What is JPA? What is Hibernate? What is the relationship between them?
2. What is an ORM (Object-Relational Mapping)?
3. What is a JPA Entity? What are the rules for creating one?
4. What is @Entity, @Table, @Id, @GeneratedValue? Explain each.
5. What are the GenerationType strategies: AUTO, IDENTITY, SEQUENCE, TABLE?
6. What is @Column? What attributes does it have (name, nullable, unique, length, precision)?
7. What is @Transient? When do you use it?
8. What is @Enumerated? Explain EnumType.STRING vs EnumType.ORDINAL and the risks.
9. What is @Temporal? When is it used (deprecated in newer Hibernate)?
10. What is @Lob? When would you use it for a TEXT or BLOB column?
11. What is @Version? What is optimistic locking?
12. What is pessimistic locking? What are LockModeType values?
13. What is @Embedded and @Embeddable? Give an example with Address class.
14. What is @AttributeOverride? When would you need it?
15. What is a composite primary key? How do you implement it — @IdClass vs @EmbeddedId?
16. What is @CreationTimestamp and @UpdateTimestamp (Hibernate-specific)?

JPA Relationships

1. What is @ManyToOne? What is its default fetch type?
2. What is @OneToMany? What is its default fetch type? What is mappedBy?
3. What is @OneToOne? What are the strategies for its mapping?
4. What is @ManyToMany? How do you customise the join table?
5. What is the owning side and inverse side of a relationship?
6. What is CascadeType? Explain all values: PERSIST, MERGE, REMOVE, REFRESH, DETACH, ALL.
7. What is orphanRemoval=true? How is it different from CascadeType.REMOVE?
8. What is FetchType.EAGER vs FetchType.LAZY?
9. What is the N+1 select problem? Give a clear example.
10. How do you fix the N+1 problem? Explain JOIN FETCH, @EntityGraph, and @BatchSize.
11. What is a bidirectional relationship? How do you keep both sides in sync?
12. What is an @JoinColumn? What FK column does it create?
13. What is @JoinTable? When is it used?
14. What is @MapsId? How is it used for shared primary keys?

Spring Data JPA — Repositories & Queries

1. What is Spring Data JPA? What does extending JpaRepository give you?
2. What is the difference between JpaRepository, CrudRepository, and PagingAndSortingRepository?
3. What is a derived query method? How does Spring Data parse the method name?
4. What are the keywords used in derived queries: By, And, Or, Not, Between, Like, In, OrderBy?
5. What is the difference between findBy and getBy and readBy in derived queries?
6. What is @Query? How do you write JPQL queries?
7. What is a named parameter in @Query? How is it used with @Param?

8. What is a native query in @Query? When would you use nativeQuery=true?
9. What is @Modifying? When is it required? What must accompany it?
10. What is Pageable? How do you implement pagination and sorting?
11. What is Page vs Slice? What is the difference?
12. What is Sort in Spring Data? How do you create dynamic sorting?
13. What is a projection in Spring Data JPA? What are interface-based and class-based projections?
14. What is @EntityGraph? How does it solve the N+1 problem?
15. What is the difference between findById() and findOne() / getReferenceById()?
16. What is save() vs saveAndFlush() vs saveAll()?
17. What is deleteById() vs delete()? What is deleteAll() and deleteAllInBatch()?
18. What is existsById()? What SQL does it generate?
19. What is the Criteria API? When would you use it over JPQL?
20. What is Specifications in Spring Data JPA? How do you build dynamic queries?

Hibernate & JPA Internals

1. What is the Hibernate Session and how does it relate to the JPA EntityManager?
2. What are the entity states in JPA: Transient, Managed, Detached, Removed?
3. What is the persistence context? What is its lifecycle?
4. What is the first-level cache (session cache) in Hibernate?
5. What is the second-level cache in Hibernate? What providers are supported?
6. What is the query cache in Hibernate?
7. What is dirty checking in Hibernate? When does a flush happen?
8. What is FlushMode? What are ALWAYS, AUTO, COMMIT, MANUAL?
9. What is the difference between merge() and persist() in JPA?
10. What is detach() and refresh()?
11. What is a LazyInitializationException? When does it occur and how do you fix it?
12. What is the Open Session in View anti-pattern?
13. What is hibernate.ddl-auto? Explain all 5 options.
14. What is the difference between update and validate in ddl-auto?
15. What is Flyway? What is Liquibase? How do they differ from ddl-auto?
16. What is the @Fetch annotation (Hibernate)? What is FetchMode.JOIN, SELECT, SUBSELECT?

Spring Security Architecture

1. What is Spring Security? What does it protect against?
2. What is the Security Filter Chain? How does it work?
3. What is DelegatingFilterProxy? How does it bridge Spring Security and the servlet container?
4. What is FilterChainProxy?
5. What is the order of key filters in the default Spring Security filter chain?
6. What is SecurityContext and SecurityContextHolder?
7. What is ThreadLocal? How does SecurityContextHolder use it?
8. What is Authentication? What is an AuthenticationManager?
9. What is ProviderManager? What is AuthenticationProvider?
10. What is DaoAuthenticationProvider?
11. What is UserDetails? What is UserDetailsService?
12. What is UsernamePasswordAuthenticationToken?
13. What is GrantedAuthority? What is the difference between a role and an authority?
14. What is the ROLE_ prefix convention in Spring Security?
15. What is SecurityContextPersistenceFilter?

JWT Authentication

1. What is JWT (JSON Web Token)? What problem does it solve?
2. What are the 3 parts of a JWT: header, payload, signature?
3. What claims are in a JWT payload? What are registered, public, and private claims?
4. What is the difference between HS256 and RS256 signing algorithms?
5. What is a JWT secret key? What makes a good secret key?
6. How do you validate a JWT token — signature check, expiry, and claim extraction?
7. What is stateless authentication? Why is JWT suitable for it?
8. What is a refresh token? How is it used to renew an access token?
9. What is the difference between session-based and token-based authentication?
10. What is OncePerRequestFilter? Why must JwtFilter extend it?
11. How do you extract the Bearer token from the Authorization header?
12. What is the flow: login request → JWT generation → JWT validation on subsequent requests?
13. How do you store JWT on the client side — localStorage vs httpOnly cookie? Pros/cons?
14. What is token expiry? How do you handle expired tokens on the client side?
15. What is jjwt library? What are its main classes?

Authorisation, CSRF, CORS & OAuth2

1. What is the difference between Authentication and Authorisation?
2. What is URL-based authorisation in Spring Security? How does it work?
3. What is method-level security? What is @EnableMethodSecurity?
4. What is @PreAuthorize? Give 3 different examples of expressions.
5. What is @PostAuthorize? When would you use it?
6. What is @Secured? How is it different from @PreAuthorize?
7. What is @RolesAllowed (JSR-250)? How do you enable it?

8. What is CSRF (Cross-Site Request Forgery)? How does Spring protect against it?
9. Why is CSRF disabled for stateless REST APIs using JWT?
10. What is CORS? Why do browsers enforce it?
11. How do you configure global CORS in Spring Security?
12. What is SessionCreationPolicy.STATELESS? What does it prevent?
13. What is BCryptPasswordEncoder? What is the strength/cost parameter?
14. What is PasswordEncoder? What other implementations are there?
15. What is OAuth 2.0? What are its grant types?
16. What is OIDC (OpenID Connect)? How does it extend OAuth 2.0?
17. What is Spring Security OAuth2 Client? What is Spring Authorization Server?

AOP Concepts

1. What is AOP (Aspect-Oriented Programming)? What problem does it solve?
2. What is a cross-cutting concern? Give 5 real-world examples.
3. What is an Aspect in AOP?
4. What is a JoinPoint? What JoinPoints does Spring AOP support?
5. What is a Pointcut? What is a pointcut expression?
6. What is an Advice? What are the 5 types of advice in Spring AOP?
7. What is a Weaving? What are the 3 types of weaving?
8. What is the difference between Spring AOP and AspectJ?
9. What are the limitations of Spring AOP?
10. What is a proxy in Spring AOP? What is the difference between JDK dynamic proxy and CGLIB proxy?
11. When does Spring use JDK proxy vs CGLIB proxy?
12. What is the self-invocation problem in Spring AOP? How do you solve it?
13. What is `@EnableAspectJAutoProxy`?
14. What is `@Aspect`? What makes a class an Aspect?

Pointcut Expressions & Advice Types

1. What is the `execution()` pointcut expression syntax? Break down each part.
2. What is the `within()` pointcut expression?
3. What is the `@annotation()` pointcut? Give an example use case.
4. What is the `bean()` pointcut expression?
5. How do you combine pointcut expressions using `&&`, `||`, and `!?`
6. What is `@Pointcut`? How do you reuse a pointcut across multiple advice methods?
7. What is `@Before` advice? What can it do and what can it NOT do?
8. What is `@After` advice? When does it run?
9. What is `@AfterReturning` advice? How do you access the return value?
10. What is `@AfterThrowing` advice? How do you access the thrown exception?
11. What is `@Around` advice? What is `ProceedingJoinPoint`? How do you call `proceed()`?
12. What is the order of advice execution when multiple aspects apply to the same method?
13. What is `@Order`? How do you control the execution order of multiple aspects?
14. What is `JoinPoint.getSignature()`? What information does it provide?
15. What is `JoinPoint.getArgs()`? How do you log method arguments in `@Before`?
16. How do you implement a performance logging aspect using `@Around`?
17. How do you implement a retry aspect using `@Around`?
18. What is an introduction (inter-type declaration) in AspectJ?

JUnit 5

1. What is JUnit 5? What are its 3 modules: JUnit Platform, JUnit Jupiter, JUnit Vintage?
2. What is the difference between JUnit 4 and JUnit 5?
3. What is `@Test`? What is `@DisplayName`?
4. What is `@BeforeEach`, `@AfterEach`, `@BeforeAll`, `@AfterAll`?
5. What is `@Disabled`? What is `@Ignore` in JUnit 4?
6. What is `@ParameterizedTest`? What are the value sources?
7. What is `@ValueSource`, `@CsvSource`, `@MethodSource` in parameterized tests?
8. What is `@RepeatedTest`?
9. What is `@Nested`? When would you use nested test classes?
10. What is `@TempDir`? When is it useful?
11. What is `assertThrows()`? How do you assert an exception type and message?
12. What is `assertAll()`? Why is it better than multiple separate assertions?
13. What is `assumeTrue()` and `assumeFalse()`?
14. What is `@Timeout`? How do you set a time limit on a test?
15. What is `@Tag`? How do you use it to filter test execution?
16. What is `@ExtendWith`? What extensions does Spring provide?
17. What is the order of test method execution in JUnit 5?
18. What is `@TestMethodOrder`? What are the ordering strategies?

Mockito

1. What is Mockito? What problem does it solve?
2. What is the difference between a mock, stub, spy, and dummy?
3. What is `@Mock` vs `@InjectMocks`? How does Mockito inject mocks?
4. What is `@Spy`? When would you use it over `@Mock`?
5. What is `@Captor` / `ArgumentCaptor`? When is it useful?
6. What is `when().thenReturn()`? What is `when().thenThrow()`?
7. What is `doReturn().when()` and how is it different from `when().thenReturn()`?
8. What is `doNothing().when()`? When is it needed?
9. What is `doThrow().when()`? When do you need it vs `when().thenThrow()`?
10. What is `verify()`? What is `times()`, `never()`, `atLeast()`, `atMost()`?
11. What is `inOrder()`? How do you verify the order of method calls?
12. What is `ArgumentMatchers.any()`, `anyString()`, `anyInt()`, `eq()`?
13. What is `@ExtendWith(MockitoExtension.class)`?
14. What is `MockitoAnnotations.openMocks(this)`? When was it needed?
15. What is `@MockBean` vs `@Mock`? When do you use each?
16. What is `@SpyBean`? How does it differ from `@Spy`?
17. What is `Mockito.reset(mock)`? Is it a good practice?
18. What is `mockStatic()`? How do you mock static methods in Mockito 3.4+?

Spring Boot Testing

1. What is `@SpringBootTest`? What does it load?

2. What is webEnvironment in @SpringBootTest? What are the options?
3. What is TestRestTemplate? How is it different from RestTemplate?
4. What is WebTestClient? When would you use it?
5. What is @WebMvcTest? What does it auto-configure?
6. What is MockMvc? How do you perform a GET request and assert the response?
7. What is @AutoConfigureMockMvc?
8. What is MockMvcRequestBuilders and MockMvcResultMatchers?
9. What is @DataJpaTest? What does it load? What database does it use?
10. What is @DataJpaTest with a real database (replace = NONE)?
11. What is @WebFluxTest?
12. What is @RestClientTest?
13. What is @MockBean in Spring Boot test context?
14. What is @TestConfiguration? How is it different from @Configuration?
15. What is @Sql and @SqlGroup? How do you pre-populate test data?
16. What is Testcontainers? How do you use it with Spring Boot for integration tests?
17. What is @DirtiesContext? When do you need it?
18. What is the difference between unit test, integration test, and end-to-end test?
19. What is code coverage? What is JaCoCo? How do you generate a coverage report?

Microservices Architecture

1. What is microservices architecture? How does it differ from monolithic architecture?
2. What are the advantages of microservices?
3. What are the challenges and disadvantages of microservices?
4. What is the 12-factor app methodology? Name and explain all 12 factors.
5. What is service decomposition? What is the strangler fig pattern?
6. What is a bounded context in DDD (Domain-Driven Design)?
7. What is an API Gateway? What responsibilities does it have?
8. What is service mesh? What is Istio?
9. What is the difference between synchronous and asynchronous inter-service communication?
10. What is REST vs gRPC vs message queues for service communication?
11. What is eventual consistency? How does it differ from strong consistency?
12. What is a distributed transaction? What is the Saga pattern?
13. What is CQRS (Command Query Responsibility Segregation)?
14. What is Event Sourcing?
15. What is the Outbox pattern in microservices?

Spring Cloud Components

1. What is Spring Cloud? What problems does it solve?
2. What is Spring Cloud Netflix Eureka? How does service registration work?
3. What is Eureka Server? What is Eureka Client?
4. What is Spring Cloud Config? What is a config server?
5. How does a microservice bootstrap its configuration from a config server?
6. What is Spring Cloud Gateway? How does it differ from Netflix Zuul?
7. What is a route in Spring Cloud Gateway?
8. What is a predicate in Spring Cloud Gateway?
9. What is a filter in Spring Cloud Gateway?
10. What is OpenFeign? How do you declare a Feign client?
11. What is the difference between RestTemplate, WebClient, and Feign?
12. What is load balancing in Spring Cloud? What is Spring Cloud LoadBalancer?
13. What is Ribbon (deprecated)? What replaced it?
14. What is Spring Cloud Sleuth? What is distributed tracing?
15. What is Zipkin? How does it visualise traces?
16. What is the correlation ID / trace ID concept in distributed systems?
17. What is Spring Cloud Stream? What is a message binder?
18. What is Spring Cloud Bus?

Resilience & Circuit Breaker

1. What is the Circuit Breaker pattern? What are its 3 states: CLOSED, OPEN, HALF-OPEN?
2. What is Resilience4j? What modules does it provide?
3. What is @CircuitBreaker in Resilience4j?
4. What is @Retry in Resilience4j?

5. What is `@RateLimiter` in Resilience4j?
6. What is `@Bulkhead` in Resilience4j?
7. What is `@TimeLimiter` in Resilience4j?
8. What is a fallback method? How do you define one with Resilience4j?
9. What is the difference between Hystrix (deprecated) and Resilience4j?
10. What is the timeout pattern in microservices?

Docker

1. What is Docker? What problem does it solve?
2. What is the difference between a Docker image and a Docker container?
3. What is a Dockerfile? What instructions does it support?
4. What is FROM in a Dockerfile? What is a base image?
5. What is the difference between COPY and ADD in Dockerfile?
6. What is the difference between CMD and ENTRYPOINT in Dockerfile?
7. What is EXPOSE in Dockerfile? What is the difference between EXPOSE and -p in docker run?
8. What is a Docker layer? How does layer caching work?
9. What is a multi-stage build in Docker? Why is it used?
10. What is docker build? What is a build context?
11. What is docker run? What are key flags: -p, -e, -v, -d, --name?
12. What is docker-compose? What does a docker-compose.yml file define?
13. What is a Docker volume? Why is it needed for a database container?
14. What is a Docker network? What are bridge, host, and overlay networks?
15. What is Docker Hub? What is a private registry?
16. What is docker exec? How do you get a shell inside a running container?
17. What is docker logs? How do you tail container logs?
18. What is the difference between docker stop and docker kill?
19. What is a Docker health check?

Kubernetes Basics

1. What is Kubernetes (K8s)? What problem does it solve over Docker alone?
2. What is a Pod in Kubernetes?
3. What is a Deployment in Kubernetes?
4. What is a Service in Kubernetes? What are ClusterIP, NodePort, LoadBalancer?
5. What is a ConfigMap? What is a Secret?
6. What is a Namespace in Kubernetes?
7. What is horizontal pod autoscaling (HPA)?
8. What is a liveness probe and readiness probe?
9. What is kubectl? What are the most common commands?
10. What is a Helm chart?

Maven, Git & CI/CD

1. What is Maven? What is POM.xml?
2. What are the Maven build lifecycle phases?
3. What is the difference between compile, test, package, install, and deploy phases?
4. What is a Maven dependency scope? Name all scopes.
5. What is the difference between SNAPSHOT and RELEASE versions in Maven?
6. What is a Maven plugin? Name 5 commonly used plugins.
7. What is a Maven repository: local, central, remote?
8. What is a BOM (Bill of Materials) in Maven?

9. What is Gradle? How does it differ from Maven?
10. What is Git? What is the difference between Git and GitHub?
11. What is git init, git clone, git add, git commit, git push, git pull?
12. What is a branch in Git? What is git checkout -b?
13. What is git merge vs git rebase?
14. What is a merge conflict? How do you resolve it?
15. What is git stash?
16. What is a pull request (PR) / merge request (MR)?
17. What is CI/CD? What is Jenkins? What is GitHub Actions?
18. What is SonarQube? What metrics does it measure?
19. What is a code quality gate in SonarQube?

Creational Patterns

1. What is the Singleton pattern? When should you use it and what are its pitfalls?
2. How do you implement a thread-safe Singleton in Java? Name 3 approaches.
3. What is the double-checked locking pattern in Singleton? What is the volatile keyword's role?
4. What is the Singleton scope in Spring? How is it different from the Java Singleton pattern?
5. What is the Factory Method pattern? How does it achieve OCP?
6. What is the Abstract Factory pattern? How does it differ from Factory Method?
7. What is the Builder pattern? What problem does it solve?
8. What is the telescoping constructor anti-pattern? How does Builder solve it?
9. What is the Prototype pattern? What is the difference between shallow and deep copy?
10. How does `Object.clone()` work? What are its limitations?
11. What is the difference between Factory Method, Abstract Factory, and Builder?
12. What is the Object Pool pattern? Where is it used in Spring?
13. What is the Dependency Injection pattern? How does it enable DIP?

Structural Patterns

1. What is the Adapter pattern? What is the difference between class adapter and object adapter?
2. What is the Bridge pattern? How does it differ from Adapter?
3. What is the Composite pattern? What is the component-leaf-composite structure?
4. What is the Decorator pattern? How is it different from inheritance?
5. What is the Facade pattern? What is the difference between Facade and Adapter?
6. What is the Flyweight pattern? What is intrinsic vs extrinsic state?
7. What is the Proxy pattern? What are the 4 types: virtual, remote, protection, smart reference?
8. What is the difference between Proxy and Decorator patterns?
9. What is the difference between Proxy and Facade patterns?
10. How does Spring AOP implement the Proxy pattern?
11. How does `@Transactional` use the Decorator pattern?
12. What structural pattern do DTOs represent in Spring?

Behavioural Patterns

1. What is the Chain of Responsibility pattern? Where does Spring use it?
2. What is the Command pattern? How does it support undo operations?
3. What is the Interpreter pattern? What is an example in Java?
4. What is the Iterator pattern? How does Java's `Iterable` interface implement it?
5. What is the Mediator pattern? What Spring component implements it?
6. What is the Memento pattern? What is a real-world example?
7. What is the Observer pattern? How does Spring Events implement it?
8. What is the State pattern? How do entity status fields implement it?
9. What is the Strategy pattern? How does Spring Security use it?
10. What is the Template Method pattern? How does `JdbcTemplate` implement it?
11. What is the Visitor pattern? What are its pros and cons?
12. What is the Null Object pattern? How does Optional relate to it?

13. What is the Repository pattern? How does Spring Data implement it?
14. What is the difference between Strategy and State patterns?
15. What is the difference between Observer and Mediator patterns?
16. What is the MVC pattern? How does Spring MVC implement it?

Single Responsibility Principle (SRP)

1. What is the Single Responsibility Principle? State it in your own words.
2. What is a 'reason to change'? How do you identify multiple responsibilities in a class?
3. Give a real-world example of an SRP violation in a Spring service class.
4. How does SRP relate to cohesion and coupling?
5. How does separating services (BillingService, NotificationService, AuditLogService) follow SRP?
6. Can a class with only one method violate SRP?
7. What is a God Class / Blob anti-pattern? How does it violate SRP?
8. How does SRP make code easier to test?
9. How is SRP different from the Separation of Concerns principle?

Open/Closed Principle (OCP)

1. What is the Open/Closed Principle? What does open for extension and closed for modification mean?
2. How is OCP achieved in Java — what language features support it?
3. Give an example where adding if-else violates OCP. How do you refactor it?
4. How does the Strategy pattern support OCP?
5. How do Spring Boot auto-configuration and @Conditional annotations follow OCP?
6. What is the 'protected variation' principle and how does it relate to OCP?
7. How does adding new @ExceptionHandler methods without changing existing ones follow OCP?

Liskov Substitution Principle (LSP)

1. What is the Liskov Substitution Principle? State the formal definition.
2. What is the classic Square-Rectangle LSP violation? Why is it a violation?
3. What are the contract rules for LSP: preconditions, postconditions, invariants?
4. How do you detect an LSP violation in code?
5. How does polymorphism relate to LSP?
6. What is the difference between LSP and method overriding rules?
7. How does Spring's @RestControllerAdvice follow LSP for custom exceptions?

Interface Segregation Principle (ISP)

1. What is the Interface Segregation Principle?
2. What is a fat interface? How do you identify and fix one?
3. Give an example of an ISP violation with a UserService interface.
4. How does Spring Data JPA's CrudRepository, PagingAndSortingRepository, JpaRepository hierarchy follow ISP?
5. What is the difference between ISP and SRP?
6. How do you decide how to segregate an interface into smaller ones?

Dependency Inversion Principle (DIP)

1. What is the Dependency Inversion Principle? State both rules.
2. What is the difference between DIP and Dependency Injection?
3. Give an example of a DIP violation and how to fix it with interfaces.
4. How does Spring's IoC container implement DIP?
5. Why does DIP make code more testable?

6. What is the Service Locator pattern? Is it an alternative to DIP? What are its drawbacks?
7. How does `@Autowired` on an interface (not a concrete class) demonstrate DIP?
8. What is the relationship between DIP and the Repository pattern?

SQL Fundamentals

1. What is the difference between DDL, DML, DQL, DCL, and TCL?
2. What is the difference between DELETE, TRUNCATE, and DROP?
3. What is a PRIMARY KEY vs UNIQUE KEY vs FOREIGN KEY?
4. What is a NOT NULL constraint?
5. What is a CHECK constraint?
6. What is the difference between WHERE and HAVING clauses?
7. What is GROUP BY? What is its relationship with aggregate functions?
8. What are aggregate functions: COUNT, SUM, AVG, MIN, MAX?
9. What is the difference between UNION and UNION ALL?
10. What is the difference between INTERSECT and EXCEPT/MINUS?
11. What is a subquery? What is a correlated subquery?
12. What is the difference between IN, EXISTS, and = in subqueries?
13. What is a Common Table Expression (CTE)?
14. What is a window function? Give examples: ROW_NUMBER, RANK, DENSE_RANK, LEAD, LAG.
15. What is CASE WHEN in SQL? Give an example.
16. What is the difference between CHAR and VARCHAR?
17. What is the difference between DATE, DATETIME, TIMESTAMP, and TIME in MySQL?
18. What is COALESCE()? What is NULLIF()?
19. What is a self-join? Give a real-world example.

Joins & Indexing

1. What are the types of JOINS? Explain INNER, LEFT, RIGHT, FULL OUTER, CROSS, SELF.
2. What is the difference between LEFT JOIN and RIGHT JOIN?
3. What is the difference between INNER JOIN and CROSS JOIN?
4. What is a NATURAL JOIN?
5. What is an index? How does it improve query performance?
6. What is a clustered index vs a non-clustered index?
7. What is a composite index? What is the left-prefix rule?
8. What is a covering index?
9. When should you NOT add an index?
10. What is EXPLAIN in MySQL? How do you use it to analyse query performance?
11. What is query optimisation? What are common techniques?
12. What is a full-table scan? When does MySQL use it?

Transactions & ACID Properties

1. What is a database transaction?
2. What are the ACID properties? Explain each.
3. What is isolation? What are the 4 isolation levels in SQL?
4. What are dirty reads, non-repeatable reads, and phantom reads?
5. What isolation level prevents phantom reads?
6. What is a deadlock in a database? How do you detect and resolve it?

7. What is a savepoint in a transaction?
8. What is the difference between COMMIT and ROLLBACK?
9. What is optimistic locking vs pessimistic locking?
10. What is MVCC (Multi-Version Concurrency Control)?

Database Design & Normalisation

1. What is database normalisation? What is the purpose?
2. What is 1NF (First Normal Form)?
3. What is 2NF (Second Normal Form)?
4. What is 3NF (Third Normal Form)?
5. What is BCNF (Boyce-Codd Normal Form)?
6. What is denormalisation? When is it appropriate?
7. What is an ER diagram?
8. What is the difference between one-to-one, one-to-many, and many-to-many relationships?
9. What is a surrogate key vs a natural key?
10. What is sharding? What is replication in databases?
11. What is connection pooling? What is HikariCP?
12. What is the difference between a relational database and a NoSQL database?

Spring Transaction Management

1. What is programmatic vs declarative transaction management in Spring?
2. What is `@Transactional`? How does Spring implement it under the hood?
3. What is the default rollback rule for `@Transactional`?
4. How do you configure rollback for checked exceptions?
5. What is transaction propagation? Explain all 7 propagation types.
6. What is `REQUIRED` propagation? What happens if a transaction already exists?
7. What is `REQUIRES_NEW`? What happens to the outer transaction?
8. What is `NESTED` propagation? How does it use savepoints?
9. What is `MANDATORY`? What happens if no transaction exists?
10. What is `SUPPORTS`, `NOT_SUPPORTED`, `NEVER`?
11. What is transaction isolation in Spring? How do you set it on `@Transactional`?
12. What is a read-only transaction? What optimisations does it enable?
13. What is the self-invocation problem with `@Transactional`?
14. What is `PlatformTransactionManager`? Name 3 implementations.
15. What is `TransactionTemplate` for programmatic transactions?
16. What is `@TransactionalEventListener`? How is it different from `@EventListener`?

Spring Events, Caching & Scheduling

1. What is `ApplicationEventPublisher`? How do you publish a custom event?
2. What is `@EventListener`? How does it work?
3. What is `ApplicationEvent`? Is it required to extend it in Spring 4.2+?
4. What is a synchronous vs asynchronous event in Spring?
5. What is `@Async`? How do you enable it? What thread pool does it use?
6. What is `@EnableAsync`? What is the default executor?
7. What is Spring Cache Abstraction? What is `@EnableCaching`?
8. What is `@Cacheable`? How does it work?
9. What is `@CachePut`? How does it differ from `@Cacheable`?
10. What is `@CacheEvict`? What is `allEntries=true`?
11. What cache providers does Spring support?
12. What is `@Scheduled`? What is `fixedDelay` vs `fixedRate` vs `cron`?
13. What is `@EnableScheduling`?
14. What is a cron expression? Break down the format.
15. What is the difference between `fixedDelay` and `fixedRate` for scheduled tasks?

Spring WebFlux & Reactive Programming

1. What is reactive programming? What are the core concepts: Publisher, Subscriber, Subscription, Processor?
2. What is Project Reactor? What is Mono and Flux?
3. What is the difference between Mono and Flux?
4. What is Spring WebFlux? How does it differ from Spring MVC?
5. What is the event loop model? How does it differ from the thread-per-request model?
6. What is backpressure in reactive streams?

7. What is @RestController in WebFlux vs Spring MVC?
8. What is WebClient? How does it differ from RestTemplate?
9. What is R2DBC? How does it enable reactive database access?
10. When would you choose WebFlux over Spring MVC?
11. What is the Reactive Manifesto?

Spring Boot Tricky Questions

1. What is the difference between `@SpringBootApplication` and `@EnableAutoConfiguration`?
2. Can you have multiple `main()` methods in a Spring Boot application?
3. What happens if you have two beans of the same type and no `@Primary` or `@Qualifier`?
4. Can a `@Service` call another `@Service` method directly (bypassing Spring proxy)?
5. What happens if `@Transactional` is on a private method?
6. What happens if `@Transactional` calls another `@Transactional` method in the same class?
7. What is the difference between `@Component` and `@Bean`?
8. Can you use `@Autowired` on a static field? Does it work?
9. What is the difference between `@RestController` and `@Controller + @ResponseBody`?
10. Can Spring inject null? What is `required=false` in `@Autowired`?
11. What is the difference between `@RequestParam` and `@PathVariable`?
12. When would a Spring Boot application start but not serve requests?
13. What is `ApplicationRunner` vs `CommandLineRunner`? What is the difference?
14. What is the difference between `@SpringBootTest` and `@ExtendWith(SpringExtension.class)`?
15. What is auto-wiring ambiguity? How do you resolve it?

JPA & Hibernate Tricky Questions

1. What is the difference between `EntityManager.persist()` and `EntityManager.merge()`?
2. Can you call a repository method outside a transaction? What happens?
3. What is the difference between `FetchType.LAZY` and the actual SQL generated?
4. What happens when you call `save()` on an entity that already exists (has an ID)?
5. What is the difference between `@JoinColumn` in `@ManyToOne` and in `@OneToOne`?
6. Can an entity have two `@Id` annotations?
7. What is the difference between `equals()` and `hashCode()` in JPA entities? Why do you need to implement them?
8. What is the problem with using Auto-generated ID in `equals()` and `hashCode()`?
9. What happens if you delete an entity that has `@OneToMany` children without `CascadeType.REMOVE`?
10. What is the ghost object problem in JPA?
11. What is the difference between a JPQL `UPDATE` query and updating via entity?
12. What happens if two threads update the same entity simultaneously without locking?
13. Can you have a transaction that spans multiple HTTP requests?

REST & Security Tricky Questions

1. What is idempotency? Is `POST` idempotent? Is `DELETE` idempotent?
2. What is the difference between `401 Unauthorized` and `403 Forbidden`?
3. What is the difference between authentication and authorisation?
4. What happens if you disable CSRF but the application uses sessions (not JWT)?
5. Can you have two `@ExceptionHandler` methods for the same exception type in `@RestControllerAdvice`?
6. What is the difference between `@ResponseStatus` on a method vs on an exception class?
7. What is the difference between a JWT access token and a JWT refresh token?
8. What happens when a JWT token is intercepted (man-in-the-middle)? How do you prevent it?
9. Can you invalidate a JWT token before it expires? What are the strategies?

10. What is the difference between `permitAll()` and `anonymous()` in Spring Security?

General Architecture & Best Practices

1. What is the difference between horizontal and vertical scaling?
2. What is a load balancer? What algorithms does it use?
3. What is caching? What is the difference between local cache and distributed cache?
4. What is Redis? How is it used in Spring Boot?
5. What is a message queue? What is Kafka? What is RabbitMQ?
6. What is the difference between synchronous and asynchronous communication?
7. What is REST vs GraphQL vs gRPC? When would you choose each?
8. What is the difference between HTTP/1.1, HTTP/2, and HTTP/3?
9. What is WebSocket? When would you use it over REST?
10. What is rate limiting? How do you implement it in Spring Boot?
11. What is the CAP theorem?
12. What is the difference between SQL and NoSQL databases?
13. What is an API contract? What is Swagger / OpenAPI?
14. What is `@OpenAPIDefinition` and `@Operation` in Springdoc OpenAPI?
15. What is clean code? Name 5 clean code principles.

SECTION N TOTAL	QUESTION COUNT SUMMARY
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Section	Topic	Questions
S1	Core Java — Basics, OOP, Collections, Streams, Lambdas	~100
S2	Exception Handling, Generics, Multithreading	~60
S3	Spring Core — IoC, DI, Bean Lifecycle, Scopes	~50
S4	Spring Boot — Auto-config, Properties, Starters, Actuator	~60
S5	Spring MVC, REST API, Validation, CORS, Error Handling	~55
S6	Spring Data JPA, Hibernate, Entities, Relationships, JPQL	~75
S7	Spring Security, JWT, OAuth2, CSRF, Authorisation	~50
S8	Spring AOP — Aspects, Pointcuts, Advice Types	~40
S9	Testing — JUnit 5, Mockito, @SpringBootTest, MockMvc	~55
S10	Microservices, Spring Cloud, Circuit Breaker, Feign	~50
S11	Docker, Kubernetes, Maven, Git, CI/CD, DevOps	~50
S12	Design Patterns — All 23 GoF patterns	~45
S13	SOLID Principles — All 5 with violations and fixes	~35
S14	SQL, Database Design, Transactions, Normalisation	~50
S15	Advanced Spring — Transactions, Events, Caching, WebFlux	~45
S16	Rapid Fire — Tricky & Miscellaneous Questions	~55
TOTAL	All Topics Combined	~875+

HOW TO USE THIS QUESTION BANK: This document contains 875+ topic-wise questions. Start with the topic you are weakest in. Answer each question mentally or on paper before checking resources. For Spring-specific questions, always relate your answer back to your EnergyGrid project. For pattern and SOLID questions, have one EnergyGrid example ready. For tricky questions in Section 16 — these are real interview favourites. Revise @Transactional self-invocation, N+1, JPA entity states, and JWT invalidation carefully — they are asked in 80% of Spring Boot interviews.